

Research Question q Connectomics

“How does the human structural connectome achieve integration of distributed functional subnetworks despite low global connection density and spatial constraints on axonal projections?”

Background Survey b

“[...] **Community detection algorithms**¹ (e.g. Louvain, link community) **identify overlapping node clusters**² with sparse inter-connections. However, these methods **fail to capture higher-order transitive structures**³.”

Inspiration Document i^* Network Physics

“The notion of **k -clique percolation in random graphs**⁴ is introduced, where k is the size of the complete subgraphs whose large scale organizations are analytically and numerically investigated. [...] We discuss why clique percolation is a novel and efficient approach to the **identification of overlapping communities in large real networks**².”

Affordance-Lifted Construction

Rank 3

Query $R(q)$

Corpus Graph G_A

